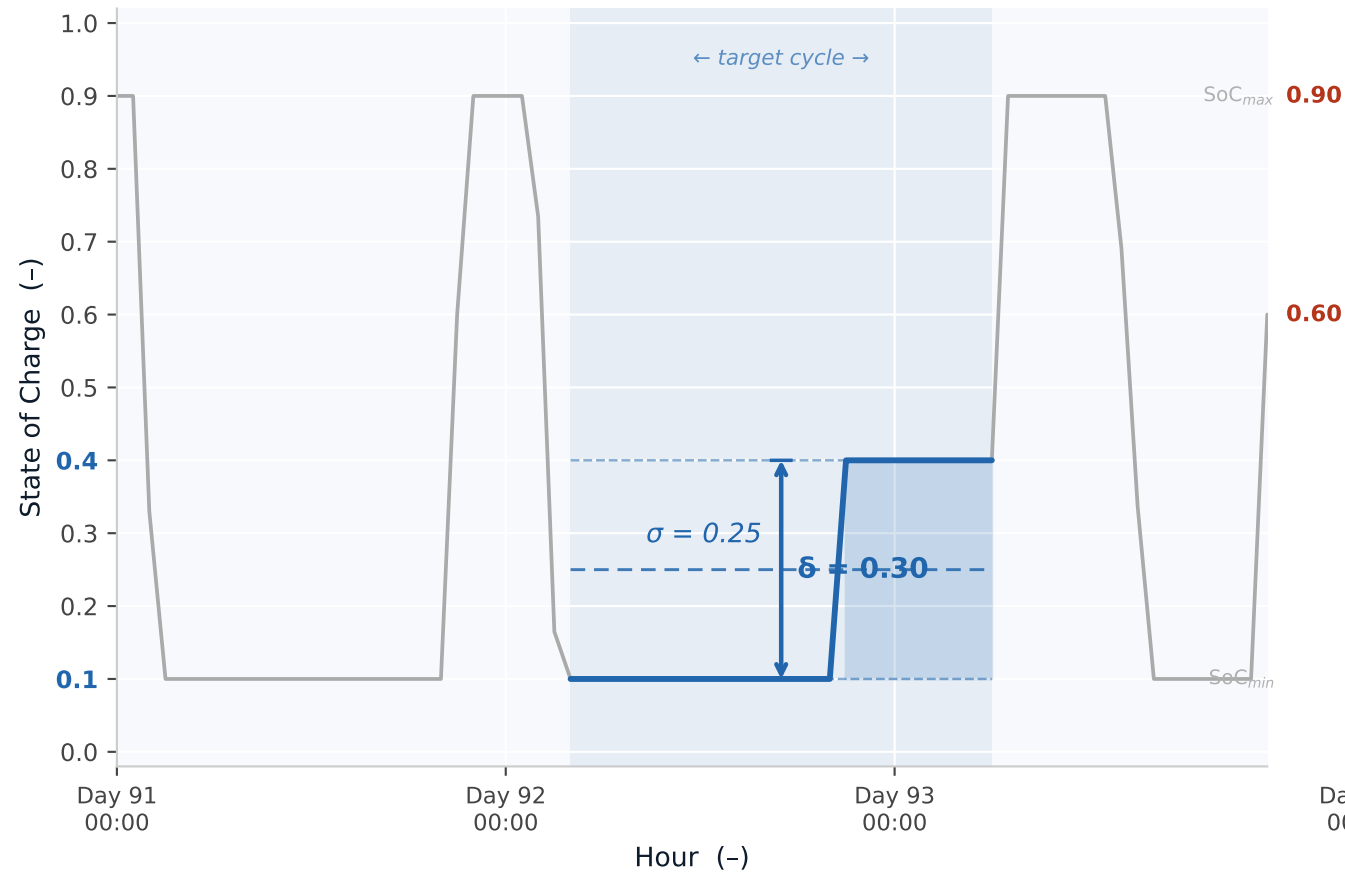


Same δ , Different σ : Same Energy Moved, Different Degradation

Both cycles: $\delta = 0.30$ (exact rainflow match, 2022 IEA Task Force 50 wind farm dispatch) S_δ identical $\rightarrow S_\sigma$ is the sole difference

Cycle A $\rightarrow$ Low σ
 $\sigma = 0.25$ window 0.10 - 0.40



Cycle B $\rightarrow$ High σ
 $\sigma = 0.75$ window 0.60 - 0.90

